SKINNER'S SCIENTIFIC SPECTRUM

"Our understanding of the basic components of nature is founded upon continued approximations that cannot be proven in every possible situation – there is no $\lim_{n\to\infty}$ for real life enabling us to rigorously prove every theory and law to an infinite degree.

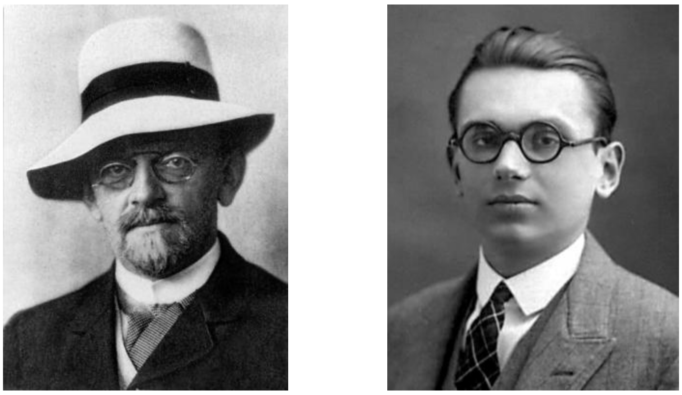
- *Neo Skinner*

Mathematics

The Cake Is a Lie!

The beauty of mathematics is that you can prove anything. Well, almost anything. In 1931 Kurt Gödel published a landmark paper in which he outlined his two incompleteness theorems that would go on to change mathematics forever. They state that any complete system of mathematics must, by definition, be incomplete. But how?

David Hilbert outlined a program to secure foundations for all of mathematics in the beginning of the 1920s when early attempts produced paradoxes and inconsistencies. This doubt within the foundations of mathematics was what Hilbert sought to remove.



Two Great Thinkers; LTR: David Hilbert, Kurt Gödel

Hilbert proposed that the foundations of mathematics should have the following qualities:

- ☞ *Formulative*: All statements can be expressed and have a precise formal language and be manipulated using well-defined rules.
- ☞ *Completeness*: All true mathematical statements can be proved.
- ☞ *Consistency*: No contradiction can be derived.
- ☞ *Conservation*: A proof using 'real objects' and 'ideal objects' can be obtained equally.
- ☞ *Decidability*: An algorithm exists for identifying truth and falsity.

For our purposes, we need only concern ourselves with formulative, completeness, consistency, and decidability - I need actually only to say, 'forget about conservation.'

To understand what Gödel said about Hilbert's program, we first need to define 'mathematics.' Mathematics is a large formal system of axioms - statements assumed to be true - with connecting rules that allows for further statements (called theorems) to be produced that are provable within the system.

Phew! That definition may sound arbitrary, but it is incredibly important as it allows us to consider elements of mathematics to prove or disprove Hilbert's proposal as Gödel himself did.

Firstly, let us consider statements written not in the language of mathematics, but in the language most familiar to us - words:

- ☞ 'This statement is *false*.'
- ☞ 'This statement is *unprovable*.'

The first of these two statements presents the idea that, if true, the statement must be false - hold on! How can that possibly make sense? Directly by proving that the first statement is true, we show it to be false. This presents a contradiction and therefore a statement that can be neither true nor false.

The first statement is an example of Gödel's second incompleteness theorem, whereby there are statements within a formal system that cannot be proved to be true or false. Hilbert's ideals of consistency and decidability are therefore violated.

But, Gödel goes one step further. In the second statement (an example of his first completeness theorem) we find that any proof of the statement is, by very definition, not a proof. This feels very similar to the first statement, but with a subtle twist: we can prove not that the statement is a contradiction, but that it is both true and unprovable. This, then, breaks the idea of completeness.

Let us now apply this to mathematics. As a formal system, we can see that the examples provided above will also apply to mathematics if only they (or equivalents) can be found within the system. Here comes the formulative property, all possible statements can be expressed within mathematics and therefore the examples above can be found and apply within the mathematical system.

Gödel showed that statements that can exist within the mathematical system can directly oppose the notion that the system is complete. If mathematics contains these statements then they cannot be proved and violate the systematic nature of Hilbert's proposal. If they do not exist within the system, then the system is incomplete. Simple.

$$G \leftrightarrow \neg \mathrm{Prov}(\ulcorner G \urcorner)$$

A Symbolic Form of Gödel's Incompleteness Theorems

Gödel's incompleteness theorems have allowed for the elegant disproval of many theories and ideas in the years since their publishing. Such examples include those of a theory of everything and logicism. Gödel's theorems have yet to be disproved and, with the continued efforts of many mathematicians, they appear to be a certain constant for any formal system of the past, present and future. Indeed, the cake as a consistent system of sponge and buttercream. Is a lie.

Article By NSR

PHYSICS

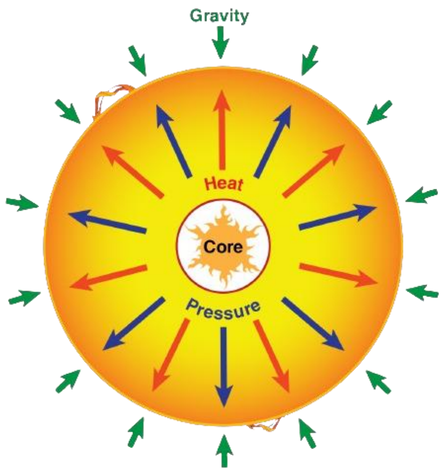
ELVIS HAS LEFT THE BUILDING.

Betelgeuse is the newest star soon to be making waves in the space. No, I am not talking about a star like Elvis, Betelgeuse is the tenth brightest star in the deep abyss of our night sky and can be seen as the top-right star in the Orion constellation. Within 100 000 years the star will experience an explosive heat death and become the second brightest object in the sky at night, behind the moon, for a period of three months.

At just 700 light years (~6.6×10^{15} km) from Earth, less than 10 million years old and 700 times larger than our Sun, Betelgeuse is a fascinating red supergiant close enough for us to measure and make predictions about but far enough away to not have any major impact on life on our small rocky home[1].

Powering every star is a large-scale thermonuclear explosion which slowly converts the hydrogen gas in the core into almost every known element; starting with helium formed from the fusion of six hydrogen atoms, a process known as nucleosynthesis pushes out hydrogen to replace the core of the star with steadily heavier elements until a small iron sphere forms at the dense centre.

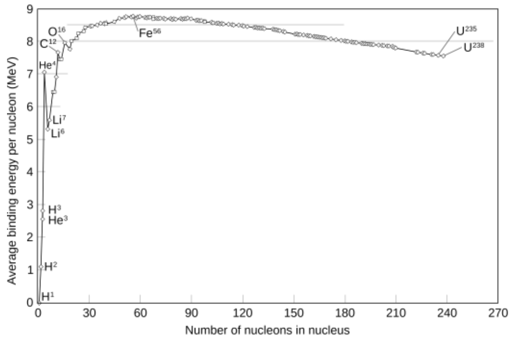
Nucleosynthesis produces large amounts of energy - a process now replicated on Earth using nuclear fusion reactors - which pushes out of the core against the force of gravity from the mass of the star itself. The battle between the force of gravity and the pressure of nucleosynthesis is what ensures the star remains in a stable balance with a semi-constant radius. This balance is similar to the repulsive electromagnetic and attractive strong nuclear forces which hold the atomic nucleus together - we will get back to this...



The Stellar Force Balancing Act

However, when iron is produced in nucleosynthesis, this careful balance is ruined and the star begins to undergo nuclear fission, producing both elements larger and smaller than iron. This imbalance leads to less pressure being exerted by the core and the star begins to collapse inwards due to its own gravity.

But why iron? Iron has the greatest binding energy and is therefore the most stable of all elements. This means that the star cannot fuse elements together past iron and loses energy due to the lower amount of energy produced by nuclear fission. Nuclear binding energy describes how tightly bound protons and neutrons are in the nucleus due to the electromagnetic and strong nuclear forces and thus how stable the atom is - there, again, is the other type of balance!

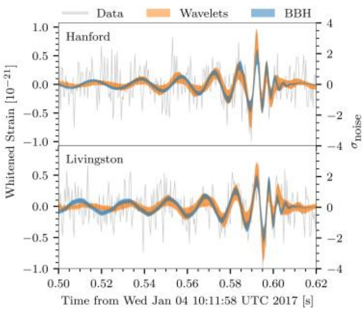


The Binding Energies of the Elements

During star death, after the star begins to collapse under gravity, there are three main options: the star forms a white dwarf, a neutron star, or a black hole. Betelgeuse, with a mass of 16.5-19 $M_\odot$ (solar masses, 1 $M_\odot \approx 2 \times 10^{30}$ kg), will decay in a supernova explosion to form a neutron star.

When Betelgeuse begins to decay, it will first produce neutrons from electron-proton collisions fuelled by gravity. These collisions also produce neutrinos which fly out of the core and travel for millions of kilometres due to their lack of interactions. The outer layers of the star are then pushed outwards by the core and produce a supernova explosion. Finally, the core and remaining outer layers implode under gravity to form a 'free core' known as a neutron star.

Supernova explosions are predicted to produce gravitational waves like those first observed at LIGO in 2016 and are the final piece of the puzzle for star death.



The Detection of Gravitational Waves at LIGO

When Betelgeuse dies, we will be able to detect it from Earth thanks to the brightness of the explosion, neutrinos produced and the gravitational waves - a final distress call from one of the largest objects in the universe.

Article By NSR

[1] Gohd

Chemistry

How To Make the Unreactive React

In 1894, Sir William Ramsay stumbled upon a component of air that was apparently unreactive. In fact, it is only the third most unreactive element known today - Argon. But, is not the third most unreactive element still unreactive? Yes, but no...

Argon was the first of the noble gases, a series of seven elements on the far right of the periodic table characterised by having a full outer (valence) electron shell which makes them traditionally completely unreactive. However, tradition is not the domain of science and modernity shall always prevail, the noble gases do indeed react, we just need a new theory of reactivity.

The first breakthrough came with noble halides - krypton hexafluoride (KrF_6), xenon hexafluoride (XeF_6) and xenon octafluoride (XeF_8) - predicted by Linus Pauling in 1933. Whilst some of these predicted compounds have been disproven and not yet experimentally observed, xenon hexafluoride has since been synthesized as a colourless solid that produces vibrant yellow vapour, exciting!

Xenon hexafluoride is formed through the reaction of elemental xenon and fluorine as follows:

$$\mathrm{Xe}\;\;\mathrm{g}\;\; + 3\mathrm{F}_2\;\;\mathrm{g}\;\; \xrightarrow{300\ ^\circ\mathrm{C}/50\ \mathrm{atm}} \mathrm{XeF}_6\,(\mathrm{s})$$

The Standard (1966) Method for Synthesis of Xenon Hexafluoride

The conditions for the reaction are the reason we do not normally find xenon hexafluoride in our everyday lives, it requires a 1:20 ratio of xenon to fluorine in a sealed container at 300 °C and 50 times the standard atmospheric pressure of Earth (101 000 Pa = 1 atm) in a reaction of 15 to 17 hours. This difficulty in synthesis is due to the unreactive nature of xenon, which is still only the fifth most unreactive of the noble gases.

Now, at this point you may well be asking why exactly noble gases can react and why it only seems to be fluorine that they react with - there is a very good reason for this. Fluorine is the most electronegative of all elements, a measure of how much the atom of an element is able to attract the pair of electrons in a covalent bond.

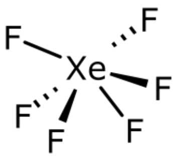
$$\chi_A = \frac{E_{ea}\;\;A\;\; + E_i(A)}{2}$$

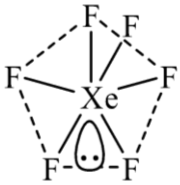
The Mulliken Scale of Electronegativity

The electronegativity of fluorine, χ_F, can be calculated using both its the electron affinity, $E_{ea}(F)$, and ionisation potential, $E_i(F)$, which are - in this case - both extremely high. Electron affinity describes how easily an element gains an additional electron whilst ionisation potential describes how difficult it is to lose an electron. The accepted value for the electronegativity of fluorine is 3.91, hydrogen has a value of only 2.25 whilst xenon is calculated at 2.73.

The high electronegativity makes fluorine a very strong oxidising agent and it is able to attract the valence electrons of xenon to form a covalent bond. In particular, it is the difference in electronegativity between fluorine and xenon ($\Delta\chi_{F,Xe} = 1.18$) that allows fluorine atoms to attract electrons from the valence shell of xenon atoms as xenon has less nuclear attraction upon the valence electrons than fluorine.

In xenon hexafluoride, six fluorine atoms form bonds with a central xenon atom and thus six out of the eight valence electrons of xenon are bound, leaving one lone pair to give a molecular shape of square bipyramidal.





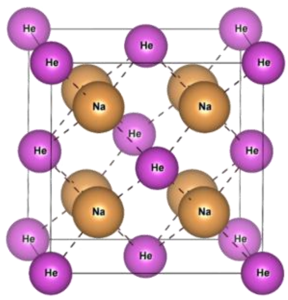
The Square Bipyramidal (Distorted Octahedral) Structure of Xenon Hexafluoride

Many more noble gas compounds have since been discovered using the elements xenon, krypton, argon, helium and even neon. This result is so surprising as neon is the most unreactive of all elements and yet forms compounds such as the ionic LiNe and covalent CF_4Ne! The trend in reactivity for the noble gases is as follows:

$$\mathrm{Ne} < \mathrm{He} < \mathrm{Ar} < \mathrm{Kr} < \mathrm{Xe} < \mathrm{Rn} < \mathrm{Og}$$

The trend generally follows the order of increasing atomic mass and radius, with the exception of helium, which is more reactive due to its electronic configuration. The p-orbitals of neon are considered to have lower forces of attraction and higher repulsions than the s-orbitals of helium.

Another noble ionic compound is disodium helide, Na_2He, which can only be formed above 100 GPa (~100 000 atm) through compressing sodium plates with helium gas to 130 GPa and heating to 1 500 K (~1 200 °C) using a laser beam. This process forces helium atoms to be reduced by the sodium atoms and form an ionic lattice.



The Crystal Lattice of Disodium Helide

Whilst we certainly will not see noble compounds in daily life anytime soon, there are already some important uses for then such as the production of 5-fluorouracil - an important cancer drug - from uracil and xenon difluoride, XeF_2. More useful than you thought!

Article By NSR

Recommendations

A-Level Wider Independent Study

- **Mathematics:** The Mathematics Behind Pringles.
- **Physics:** What Causes the Aurora Borealis?
- **Chemistry:** What Are the Uses of Germanium?

Discrete Further Reading

- **Mathematics:** *Plus Magazine May 2024*, Heads, Bayes wins!
- **Physics:** *Quanta Magazine April 2024*, How a NASA Probe Solved a Scorching Solar Mystery
- **Chemistry:** *New Scientist May 2024*, We now know exactly how thick the boundary between water and air is

Continuous Further Study Topics

- **Mathematics:** Euclid's Axioms
- **Physics:** Classifications of Stars
- **Chemistry:** The Discovery of the Noble Gases

Monthly Problem

Solve for x:

$$2 = \sqrt{x\sqrt{x\sqrt{x\sqrt{x\cdots}}}}$$

Solution next month.

> **Last month's solution:**
> $x = 2028$

Problem By the Maths Masters, Suggestions By NSR

Credits

Sources & Provenance

Mathematics:

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Images:

- ☞ Eithiraj, R.D. (2023). *Crystal Structure of Na_2He.* [Digital] *Wikimedia Commons.* Available here.
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Equations:

- ☞ Equations by LaTeX
- ☞ Equations modified for publishing by Lagrida

Chemical Diagrams:

- ☞ Chemical models by MolView
- ☞ Lewis structures by ChemDoodle

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Data sources are cited within the parent article.

Editors & Writers

- ☞ **NSR:** Neo Skinner (Editor & Lead Writer)
- ☞ Your name could go here! (see below)

Acknowledgements

The editor would like to dedicate this issue, as a token of their continued gratitude, to the reader. The readers who provide corrections and feedback are especially valuable to the editor, who wishes only that this continues such that the reader informs future issues. The courage and patience of the reader will - most surely - be rewarded through detailed and informative articles.

Corrections

- ☞ **Issue 1 - Physics:** Mr T. Barrett identified that GFCI devices are primarily used in North America and that residual current devices (RCDs) and residual current circuit breakers with overcurrent protection (RCBOs) are used in the United Kingdom. RCDs and RCBOs function in a very similar way to GFCI devices but are circuit breakers positioned in consumer units (fuse boxes/distribution boards) rather than in sockets (outlets). The use of the term GFCI was intentional over RCD due to the simplicity of the North American standard. Nice spot!

Appeal

The editor is *still* looking for a biology writer willing to put time and effort into producing a monthly article and recommendations for this publication. Hopeful applicants and deranged scientists should email their particulars to the editor, Neo Skinner, at skinners-science-spectrum@magna-aspirations.org.